AMERICAN PSYCHOLOGICAL ASSOCIATION

Psychological Review

© 2024 American Psychological Association
ISSN: 0033-295X

2025, Vol. 132, No. 4, 759–780
https://doi.org/10.1037/rev0000498

Efficient Visual Representations for Learning and Decision Making

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The efficient representation of visual information is essential for learning and decision making due to the complexity and uncertainty of the world, as well as inherent constraints on the capacity of cognitive systems. We hypothesize that biological agents learn to efficiently represent visual information in a manner that balances performance across multiple potentially competing objectives. In this article, we examine two such objectives: storing information in a manner that supports accurate recollection (maximizing veridicality) and in a manner that facilitates utility-based decision making (maximizing behavioral utility). That these two objectives may be in conflict is not immediately obvious. Our hypothesis suggests that neither behavior nor representation formation can be fully understood by studying either in isolation, with information processing constraints exerting an overarching influence. Alongside this hypothesis we develop a computational model of representation formation and behavior motivated by recent methods in machine learning and neuroscience. The resulting model explains both the beneficial aspects of human visual learning, such as fast acquisition and high generalization, as well as the biases that result from information constraints. To test this model, we developed two experimental paradigms, in decision making and learning, to evaluate how well the model's predictions match human behavior. A key feature of the proposed model is that it predicts the occurrence of commonly found biases in human decision making, resulting from the desire to form efficient representations of visual information that are useful for behavioral goals in learning and decision making and optimized under an information processing constraint.

Keywords: reinforcement learning, decision making, representation learning

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All cognitive systems face constraints on the ability to store or process information. In biological agents, these limitations range from metabolic expenditure to the bandwidth of neural information processing channels. Such constraints are especially notable in visual perception due to the incredible complexity of the visual signals that biological systems must process. Due to these constraints, biological agents must form informationally simplistic representations of their sensory world that still contain information relevant to behavioral goals. This can be achieved by ignoring or abstracting away information that is irrelevant to the task at hand. However, ignoring too much information when forming representations can negatively impact our ability to perform other behavioral tasks, such as accurately recalling previous information. In addition, dynamic tasks may result in previously irrelevant information becoming more relevant for a given task, requiring more information to be represented. How then, should biological systems represent visual information in a manner that optimizes performance across a range of competing tasks?

Recent theories in cognitive science have sought to explain how the brain solves the challenges introduced by cognitive limitations by taking the biological cognitive system to be analogous to an adaptive information processing system (Bates et al., 2019; Bates & Jacobs, 2020; Bhui et al., 2021; Lai & Gershman, 2021). This approach entails the use of formal information-theoretic principles and other tools from communication engineering applied onto biological cognition. This is achieved by assuming a *cognitive communication system* that translates information across a channel with a specific capacity and optimizing for some behavioral objective given that capacity. In this work, we apply this approach to addressing the goal of optimal representation formation of visual information with respect to both cognitive constraints and behavioral goals in utility-based learning and decision making.

One relevant concept is the Information Bottleneck Approach (Tishby et al., 2000), which considers the problem of optimal communication across a channel with a finite capacity. The solution to this problem can be reached through the use of the Blahut–Arimoto

This article was published Online First September 19, 2024.
Hongjing Lu served as action editor.
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All experiment participant data, stimuli models, and code are made available on Open Science Framework at https://osf.io/h97s4/. The two experiments were not preregistered.
The authors have no conflicts of interest to disclose. This article was completed during the fulfillment of Tyler Malloy's PhD degree in Cognitive Science at Rensselaer Polytechnic Institute, advised by Chris R. Sims. This research was supported by National Science Foundation Grant DRL-1560829 (awarded to Chris R. Sims) and a grant from the RPI-IBM Artificial Intelligence Research Collaboration (awarded to Tyler Malloy).
Tyler Malloy played a lead role in conceptualization, data curation, formal analysis, investigation, methodology, software, visualization, writing–original draft, and writing–review and editing. Chris R. Sims played a lead role in funding acquisition, investigation, project administration, resources, supervision, and writing–review and editing and a supporting role in writing–original draft.
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algorithm from the field of rate–distortion theory (Arimoto, 1972; Blahut, 1972). This rate–distortion theory inspired approach has since been applied onto understanding the optimal functioning of decision making and perception (Tishby & Polani, 2011), visual working memory (VWM; C. R. Sims et al., 2012), absolute identification (C. R. Sims, 2016), and perceptual generalization (C. R. Sims, 2018).

The challenge of understanding the optimal representation of visual information also exists within the field of machine learning, where methods share similar inspiration from the information bottleneck principle. Artificial neural network (ANN) models have been integrated with cognitive models to solve various goals such as modeling the representation of natural language and visual information in utility-based tasks (Malloy & Gonzalez, 2024). In (Bates et al., 2024), the authors apply ANNs to modeling VWM in a task that required participants to recall the orientation of an artificially generated natural scene, and accurately predicted participant behavior using a large pretrained vision model provided by OpenAI. The authors additionally compared their large pretrained model to several baselines, one of which was a β-variational autoencoder (β-VAE) model, which predicted human behavior reasonably well, though to a lower accuracy than the large pretrained model. Due to the abstract nature of our task compared to the naturalistic images used in (Bates et al., 2024), we preferred the used of smaller ad hoc trained β-VAE models.

Another important quality of β-VAE models is their adjustable information bottleneck (Higgins et al., 2017). This type of ANN is trained with an objective that shares a common motivation as the information bottleneck approach, in terms of optimizing performance relative to an adaptive information processing constraint (Burgess et al., 2018; Sikka et al., 2019). In this way, β-VAE models in machine learning are related to models that take cognition to be a communication system that optimizes for some behavior under a constraint. Recent applications of β-VAEs to modeling biological cognition have also demonstrated a close correspondence to the activity of individual neurons in Macaque monkeys while they were observing images of human faces (Higgins et al., 2021). These models showed similar activity of neurons in specific brain regions associated with visual perception without being trained to do so directly. Instead, β-VAE models are trained to reduce the inaccuracy of a stimuli reconstruction, while keeping the internal representation informationally simplistic. This is done by balancing reconstruction accuracy and an information-theoretic measure of representation complexity.

To better understand β-VAE models and how they relate to our proposed model, the following sections begin with background on information theory. This will clarify the conceptualization of cognition as an information processing channel with a finite capacity. After this, there is a section that includes an in-depth description of the previously mentioned cognitive communication channel, followed by details of the β-VAE model that serves as a basis for the model presented in this article.

After these background sections, we present two novel experiment paradigms both involving visual change detection as well as either utility-based learning or decision making. The purpose of these experiments is to investigate how visual representations interact with other cognitive processes, such as learning form feedback and decision making based on learned utility values. Comparing human behavior with our proposed cognitive model allows for an analysis of the similarity of memories formed by humans and our cognitive model's representations.

Our proposed model is then described in detail, including how it is applied to predicting human behavior. Next, a comparison of our proposed model against related methods is performed, as well as an analysis of the representations formed by these models. Finally, a general discussion will provide summary and insights from the cognitive modeling of human behavior in the experiments we present.

Information Theory and Cognitive Processing

In this work, information theory will be applied to understanding human visual representation formation, learning, and decision making. The basic foundation of information theory assumes the concept of an information source which produces a message that will be communicated across an information channel (Shannon, 1948). This information source can be understood as a set of possible discrete messages X with corresponding probabilities of occurring $p(x)$, $x \in X$. This allows for a definition of the fundamental information-theoretic quantity of entropy as $H(x) = -\Sigma_{x \in X}\, p(x) \log p(x)$ which represents the uncertainty of a random variable.

Discriminability is defined as the ability to tell the difference between two or more stimuli. As the amount of information that is known about a set of stimuli is increased, the ability to distinguish between them similarly increases. The goal of discriminability in electrical communication systems is to ensure that all inputs to the communication system are transmitted to an output in a manner such that it is possible to distinguish between the inputs based on their output. This is related to task-relevant perception and memory as both processes are useful in so far as they are able to distinguish between other task-relevant perceptions and memories.

Choice is related to information through the reduction of uncertainty that results from an increase in information, which directs which option should be chosen. Discriminating between the features of choices is key to selecting between them, and failing to distinguish between these features can lead to suboptimal selections. Since humans have a finite capacity for processing information, their choice selection and decision making are similarly limited, but by attending to information that is more relevant and choosing what to ignore, their inattention to specific information can be understood as rational (C. A. Sims, 2003).

With respect to the behavior, channel capacity determines an upper bound on performance that can be achieved by a system with a given information processing ability. For example, a system with severely limited channel capacity might be incapable of making fine-grained perceptual distinctions. In turn, this lack of perceptual distinction can impact the accuracy of actions relative to behavioral goals. The achievable level of performance depends not only on the capacity of the channel, but also the complexity of the information that is being processed, as more difficult learning and decision-making problems would naturally require a higher information processing capacity.

Autoencoder Models

Autoencoders are a class of ANN models that are trained to learn a sparse encoding of an input such that it can be reconstructed with as little error as possible. These models were designed to solve the *encoding problem* of representing an input that contains N bits of information, using a smaller number of bits as a pattern that can be

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used to reconstruct the original input (Barlow, 2012) often with some loss to the original input. This is directly related to the challenge of communication across a noisy information channel, with the capacity of the channel serving as an analogy for the representation information constraint.

To achieve this, autoencoders use an encoder that forms low-dimensional representations of inputs, and a decoder that reconstructs inputs based on these representations. One class of ANN methods that solves this encoding problem are called variational autoencoders (VAEs), which learn an encoder model $q_\phi(z|x)$ that forms representations z that are samples from an N-dimensional multivariate Gaussian distribution, $z \sim N(M, \Sigma)$, where the covariate matrix Σ is diagonal with values $[\sigma_1^2, \sigma_2^2, \ldots, \sigma_n^2]$ and the vector of means is $M = [\mu_1, \mu_2, \ldots, \mu_n]$ (Kingma & Welling, 2022). The sampled representations z (also called latent samples) are fed through decoder layers $p_\theta(x|z)$ to produce a reconstruction of the original signal. This model is trained by maximizing the following objective (Higgins et al., 2017):

$$\mathcal{F}(\theta, \phi; x, z) = \mathbb{E}_{q_\phi(z|x)}[\log p_\theta(x|z)] - D_{\mathrm{KL}}(q_\phi(z|x)\|p(z)). \quad (1)$$

In Equation 1, ϕ represents the parameters of the encoder $q_\phi(z|x)$, which defines the probability of the encoder producing the representation z based on the input x. The decoder is parameterized by θ and defined by the probability $p_\theta(x|z)$ of observing the output x given the internal representation z. The final term $D_{\mathrm{KL}}(q_\phi(z|x)\|p(z)) = \sum_{x \in X} q_\phi(z|x) \log \frac{p(z)}{q_\phi(z|x)}$ is the Kullback–Leibler-Divergence D_{KL} between the probability of the decoder $q_\phi(z|x)$ and the marginal probability of the representation $p(z)$; this term corresponds to the informational complexity of the representations (Burgess et al., 2018).

The goal of the training objective is to maximize the expectation $\mathbb{E}$ of the log of the decoder probability $p_\theta(x|z)$. This is achieved by learning a decoder where $p(x|z) \approx p(x|v, w)$ where v are the conditionally independent factors responsible for generating the input x, and w are the conditionally dependent factors. This distinction between conditionally independent and dependent factors is how the model is able to learn an encoder and decoder that uses representations with less information than the original data source (Higgins et al., 2017). This is achieved by abstracting away the redundant information contained in the conditionally independent factors, and learning a representation that is sensitive to conditionally dependent factors. The section on the proposed model gives an example which is used to describe the difference between conditionally dependent and independent factors.

An alteration to the VAE modeling approach is the β-VAE (Higgins et al., 2017). These models use an additional parameter β that is used to additionally control the information bottleneck of the model, beyond the dimensionality of the representation. This β parameter is included in the maximization training objective of the β-VAE model:

$$\mathcal{F}(\theta, \phi; x, z, \beta) = \mathbb{E}_{q_\phi(z|x)}[\log p_\theta(x|z)] - \beta D_{\mathrm{KL}}(q_\phi(z|x)\|p(z)). \quad (2)$$

The impact on training of this additional term is to control the informational complexity of the learned representations of the β-VAE model. Increasing the value of the parameter β induces a more constrained information bottleneck by reducing the informational capacity of the model representations, and decreasing this value allows for more complex representations.

A result of controlling this information bottleneck is that it encourages the so-called *disentangled* representations of the input x, meaning that variations in one dimension of the features of an input result in variations along only one dimension of the representation (Burgess et al., 2018). The disentangled property of β-VAE learned representations has demonstrated a close correspondence to the activity of macaque monkey neuronal firing while the monkeys were observing images of human faces with different features (Higgins et al., 2021). The source of disentanglement in the representations of β-VAE models is the information bottleneck (Achille & Soatto, 2018) which motivates a connection between the use of these models with previous relations of cognition and communication by the application of information theoretic principles.

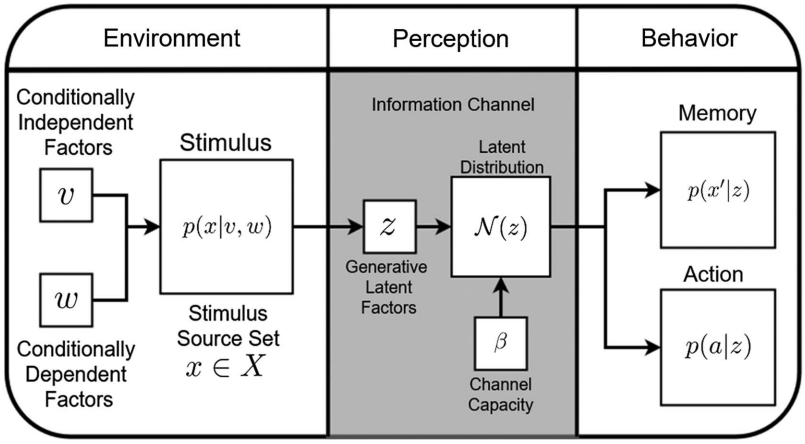
Cognitive Communication Channel

The schematic of our cognitive communication channel, shown in Figure 1, is intended to illustrate how information is translated in visual representation formation during learning and decision making. This formulation of representation formation derives from a β-VAE model, which seeks to achieve accurate reconstructions x' of original stimuli x by training the model to reduce the difference between the stimuli input and the reconstruction output, while also penalizing the informational complexity of the internal latent representation. In this figure, β represents the parameter of the β-VAE model training objective in Equation 2. $\mathcal{N}(z)$ represents the distribution $\mathcal{N}(M, \Sigma)$ defined by the latent representations of the β-VAE model z. The result is a model of perceptual representation formation including an information capacity describing the impact on both memory and action.

The environment that a cognitive agent exists in consists of the conditionally dependent factors v and conditionally independent factors w that are responsible for generating the stimulus x that the agent experiences according to the probability $p(x|v, w)$. The goal of visual representation formation is to learn a latent representation z that is useful for both memory recollection $p(x'|z)$ and task-relevant behavior $p(a|z)$. Note that these are just two possible uses for a mental representation. The broader point is that cognition must learn representations that are both informationally compact, while facilitating performance across a range of possible tasks. The optimality of this representation must also take into account the information capacity of the cognitive agent, which is formalized by restricting the informational complexity of the distribution that is sampled from to generate memory and actions. The constraint on representation complexity is represented by the parameter β which can be increased or decreased in the model to conform to a specific channel capacity.

Given an observed stimulus $x \in X$ the cognitive agent must learn the joint distribution of the stimulus and the generative latent factors z such that $p(x|z) \approx p(x|v, w)$ (Higgins et al., 2017). The difference between conditionally independent and dependent factors is crucial to understanding how a stimulus should be optimally represented. Ideally, latent factors z should be disentangled, meaning they are sensitive to individual changes in the conditionally independent factors v and not necessarily to conditionally dependent factors w (Higgins et al., 2018). Using the example of images of human faces previously discussed, these representations would retain information related to causal factors that produce differences between human faces, such as age, gender, or race, and abstract information that is

Figure 1
A Cognitive Communication Channel



Note. Schematic of the cognitive communication channel including the environment that generates stimuli, consisting of conditionally independent and dependent factors, the perceptive faculty of the cognitive agent that generates latent factors used to define an information-constrained distribution that can be sampled from to generate both memory and actions.

generally similar among images of human faces, like average head position and shape. This individual sensitivity allows the model to abstract away irrelevant information when the information capacity requires it, while preserving the accuracy of memory reconstruction $p(x'|z)$ and actions $p(a|z)$.

While the disentanglement objective can result in beneficial properties for representation learning, abstracting away any information introduces the possibility of negatively impacting memory and decision making. The goal of our model is to investigate how representations of visual information impact both memory and behavior. In the next sections, we describe in detail two common biases that are observed in human visual memory and utility-based learning and decision making. Investigating these biases and their potential causes is the motivation behind the design of the two experiments we present.

Behavioral Biases

Biased Utility Choices

When making decisions under risk and uncertainty, humans often exhibit a sensitivity to the risk associated with outcomes preferring more or less risky outcomes depending on the specific outcome probabilities of the decision being made. Preferring a choice that is associated with a larger distribution of possible outcomes is referred to as *risk aversion*, while preferring a smaller distribution is referred to as *risk seeking* (Resnik, 1987). An example risk-averse behavior would be strongly preferring a wager that pays off \$3 with a 50% chance, and \$1 with a 50% chance compared to a wager that pays off \$200 with a 1% change, and 0\$ with a 99% chance. Since the two wagers have equivalent expected values, there should not be a strong preference under expected utility theory (Resnik, 1987). The opposite preference would be an instance of risk-seeking behavior.

If utility-based choice is determined by taking the maximum expected utility alone, then selecting choices in a manner that is sensitive to the underlying risk would be seen as a deviation from rationality. However, risk aversion has been attributed to finite capacity information processing faculties (C. A. Sims, 2005), meaning that an agent that allocates attention rationally with respect to its information processing capacity may produce risk-averse behavior in some situations. One alternative explanation for the source of risk-sensitive behavior is that humans have an attentional bias that overweighs more likely outcomes, producing biases related to the risk associated with outcomes (Tversky & Kahneman, 1992). However, this account does not explain the source of this attentional bias or why it might be beneficial for humans to attend to outcomes differently depending on their likelihood.

While previous explanations have sought to understand how information processing capacities can be related to risk-sensitive behavior, they typically deal with access to information (C. A. Sims, 2005), which fails to explain how information processing constraints impact decision making based on visual information. This explanation is given by our proposed model, which understands the phenomenon of risk sensitivity in relation to the information that is contained in constrained representations of visual information associated with decision making tasks. Another important aspect of how capacity relates to behavioral biases is through the interaction between attention and the constrained representation of information in memory, which has be related to a variety of behavioral phenomenon (Radulescu et al., 2021). This work seeks to further the understanding of this relationship and its impact on behavior.

Biased Visual Memory

Visual memory can be considered biased in many ways, in this work, we focus on biases introduced by cognitive constraints and domain relevancy. In VWM research, the contents of VWM are

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generally assumed to be both information constrained (Ballard et al., 1995) and task dependent (Hayhoe et al., 1998). These two features of visual memory are intrinsically aligned, as the limitations imposed by cognitive constraints force rational agents to focus attention onto information that is relevant for a given domain to achieve their behavioral goals. The goal of this work is to describe the optimal allocation of limited informational resources when forming representations of visual information, given constraints imposed by cognitive realities and in relation to task demands. The accuracy of the proposed model will be assessed, in part, by its ability to explain biases present in human visual memory, and how well it predicts the behavior of human participants in a task that tests VWM accuracy.

Many visual biases and illusions are well studied and described by previous literature. Often, these are seen as mistakes resulting from the use of heuristics that are required to simplify real-world tasks. The model presented in this work instead views these visual memory and representation formation biases as resulting from the optimal allocation of limited informational resources in a manner analogous to an information processing channel. One particularly interesting bias that is relevant for both the proposed model and the experiment paradigm described in the next section is that of *acquired equivalence*, whereby visually dissimilar stimuli can be represented as more similar, and thus more likely to be mistaken for one another, after a participant gains experience associating them with similar behavior in the same situation (Honey & Hall, 1989). Instead of viewing this as the result of a specific heuristic, the model presented here views acquired equivalence as resulting from a representation formation strategy that balances the accuracy of recall and the usefulness for a given task, both in relation to cognitive constraints. Importantly, this three-part balance is optimized in a manner described by information theory, which motivates the model we propose.

Information Theory and Biases

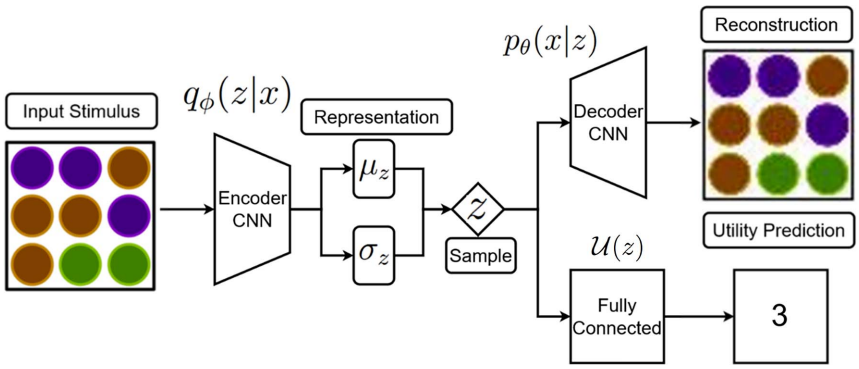
Risk-averse and risk-seeking behavior have previously been associated with the optimization of behavioral results under a constraint in terms of an informational capacity (Woodford, 2012). These phenomena have also been investigated in the decision making by sampling literature, in relation to differences in the size of outcome samples and the dynamics of the decision-making task (Bhui & Gershman, 2018). Several studies of biased decision making have found that encoding decision making problems using a lossy encoder can produce risk sensitive effects (Frydman & Jin, 2021; Khaw et al., 2021). However, these previous studies have investigated these biases in relation to the information contained in descriptions of the decision-making problems, rather than directly modeling the representation of visual information related to a task. The model proposed in this work furthers the understanding of how information constraints relate to behavioral biases by directly modeling the process of visual representation formation using our proposed alteration to the β-VAE model.

Proposed Model

The model we propose for understanding human behavior and representation formation in the two experiment tasks is the utility β-variational autoencoder model (Uβ-VAE). This is a deep neural network based model that takes as input a visual stimulus and produces an information-constrained representation used for both reconstruction and utility prediction. This architecture is similar to previous models of modeling visual memory representations (Higgins et al., 2021), visual categorization tasks (Bates & Jacobs, 2020), visual utility-based decision making (Malloy & Sims, 2021), and abstract hypothesis generation and evaluation (Malloy et al., 2022).

Figure 2 details the proposed model, the structure of which can be considered as a modification to a traditional β-VAE to additionally make predictions of utility associated with visual stimuli. Updates to the utility prediction are applied to previous layers of the network, meaning that the accuracy of utility predictions can change the way that stimuli are represented. The Uβ-VAE model is trained using a training objective which balances reconstruction error, representation complexity, and utility prediction accuracy. This utility accuracy is included by the addition of the squared error of the utility prediction and true observed utility, which is adjusted with the parameter υ

Figure 2
Proposed Model



Note. Proposed model design including encoding convolutional neural network layers, the internal representation that defines a distribution that is sampled from and fed into the decoder convolutional layers to form a lossy reconstruction. This sample is also fed into fully connected layers to produce utility predictions. CNN = convolutional neural network. See the online article for the color version of this figure.

$$\mathcal{F}(\theta, \phi; x, z, \upsilon, \beta) = E_{q_\phi(z|x)}[\log p_\theta(x|z)] - \beta D_{\mathrm{KL}}(q_\phi(z|x)\|p(z)) + \upsilon(U(z) - u(x))^2 = \mathcal{F}(\theta, \phi; x, z, \beta) + \upsilon(U(z) - u(x))^2, \quad (3)$$

which controls the relative importance of utility prediction accuracy during training. This results in the equation:

(see Equation 3 above)

where $\mathcal{F}(\theta, \phi; x, z, \beta)$ is the base β-VAE training objective shown in Equation 2. In this Uβ-VAE training objective, shown in Equation 3, U represents the utility prediction module which outputs the prediction $U(z)$ based on the latent representation z. $E_{q_\phi(z|x)}[\log p_\theta(x|z)]$ represents the reconstruction error, $\beta D_{\mathrm{KL}}(q_\phi(z|x)||p(z))$ represents the information complexity weighted by the parameter β, and $\upsilon(U(z) - u(x))^2$ represents the utility prediction error weighted by the parameter υ. The relative importance of the utility prediction accuracy is weighted by the parameter υ. To update utility prediction accuracy, the value of $u(x)$ is determined either with the ground truth stimuli mean expected utility in the case of decision making, when that information is presented to the human participant, or the direct utility outcome observed by a participant in the case of learning.

Uβ-VAE Model Representations

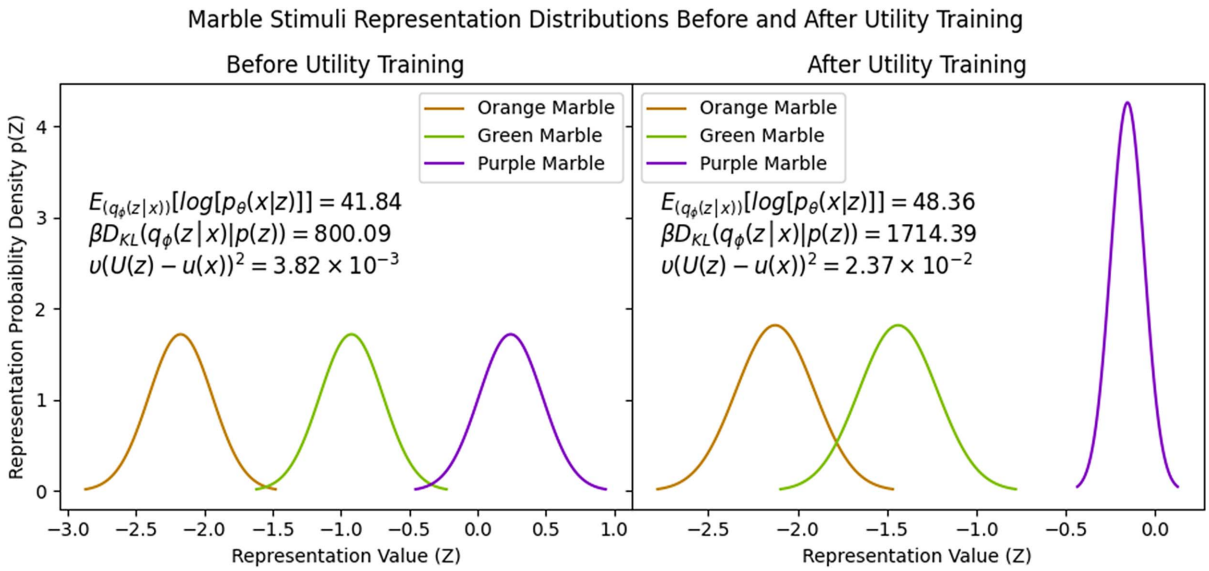
The proposed Uβ-VAE model predicts how utility impacts learned representations of visual information. This can be illustrated by constructing a Uβ-VAE model that uses just one latent dimension, allowing for a simple visualization of the learned representations, and comparing the impact of utility training. To do this, we developed a simple data set consisting of images of a marble, with the marble's color sampled randomly and uniformly from the set (green, orange, purple). Associated with each marble was a utility value. The goal for the model is to learn to encode both the visual appearance of the stimuli, along with their utilities. This example can also demonstrate the difference between conditionally dependent (w) and independent (v) factors. The independent factors are the shape of the marble stimuli (circular), and the dependent factors are the color of the marble. The encoder should not spend informational resources on representing the independent factors, since all marbles in these stimuli are the same shape, and instead use resources to represent the dependent factors of the color of the marbles.

We trained a Uβ-VAE model for 1,000 epochs to reconstruct the stimuli and predict the utility using the same utility of one-third for all marbles. This results in a representation distribution of the latent variables z on the left-hand side of Figure 3. After this, we train this model on an updated utility distribution where the orange and green marbles have a utility of 0, and the purple marble with a utility of 1. The impact of these updated utilities is shown on the right of Figure 3.

Compared to the distributions before training, we can observe an increase in the reconstruction error $E_{q_\phi(z|x)}[\log p_\theta(x|z)]$ from 41.84 to 48.36, as the distributions of the two similarly low utility stimuli are now slightly overlapping, increasing the probability of them being mistaken for one another when a sampled value of z is reconstructed. Additionally, the weighted Kullback–Leibler-divergence $\beta D_{\mathrm{KL}}(q_\phi(z|x)||p(z))$ has also increased from 800.09 to 1,714.39. The utility prediction error $\upsilon(U(z) - u(x))^2$ increased from 3.82×10^{-3} to 2.37×10^{-2}, indicating that the more informationally complex utility distribution has negatively impacted the utility-based decision making of this model.

Figure 3

Change in Marble Stimuli Representation Distributions After Training the Uβ-VAE Model on an Updated Utility Distribution



Note. $E_{q_\phi(z|x)}[\log p_\theta(x|z)]$ signifies the reconstruction error. $\beta D_{\mathrm{KL}}(q_\phi(z|x)||p(z))$ signifies the weighted informational complexity of the representations. $\upsilon(U(z) - u(x))^2$ signifies the utility prediction error. Uβ-VAE = utility β-variational autoencoder. See the online article for the color version of this figure.

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An important aspect of how utility impacts the representations learned by the Uβ-VAE model is that it has implications for how different stimuli might be mistaken for each other based on their similarity. Specifically, it suggests that stimuli that are associated with similar utilities might be mistaken for each other. This is demonstrated in Figure 3 as the distributions of the orange and green marbles are overlapping. When the orange or green representations are sampled from, there is a small chance that the sampled value will be between −2 and −1.5. Then, when this sampled value is fed back into the reconstruction layers of the proposed model, they might result in a reconstruction that does not match the original input. This illustrative example uses the simple case of a single marble but has implications for more complex stimuli. The experiments used in this work are designed to compare the predicted behavior of this proposed model and human behavioral biases.

Another effect can be observed by comparing the Uβ-VAE model after training it on these simple stimuli using different utilities associated with the purple, orange, and green marbles. This is done by using variable utility outcomes for the three marbles. For this simulation, the orange and green marbles will have lower utilities with lower variance, a normal $\mathcal{N}(2,1)$ distribution; while the purple marbles have a higher utility and higher variance, a normal $\mathcal{N}(3,2)$ distribution. Then, we can compare the probability of selecting one marble (called x_1) over an alternative (called x_2) based on the utility predictions of the Uβ-VAE model and the true outcome variance difference between the marble pairs, as shown in Figure 4.

Training the Uβ-VAE model with different β parameters produces risk-sensitive predictions of marble utilities; when the β parameter is high the predictions are risk seeking, and when the same parameter is low the predictions are risk averse. This is due to the difference in the information constraint imposed by our proposed model, which results in different errors when predicting utility, based on the relative information constraint. These model biases are closely related to behavioral biases related to risk sensitivity and acquired equivalence observed in humans, which are evaluated experimentally in the following section using more complex stimuli, before comparing with model behavior.

Learning and Decision-Making Experiments

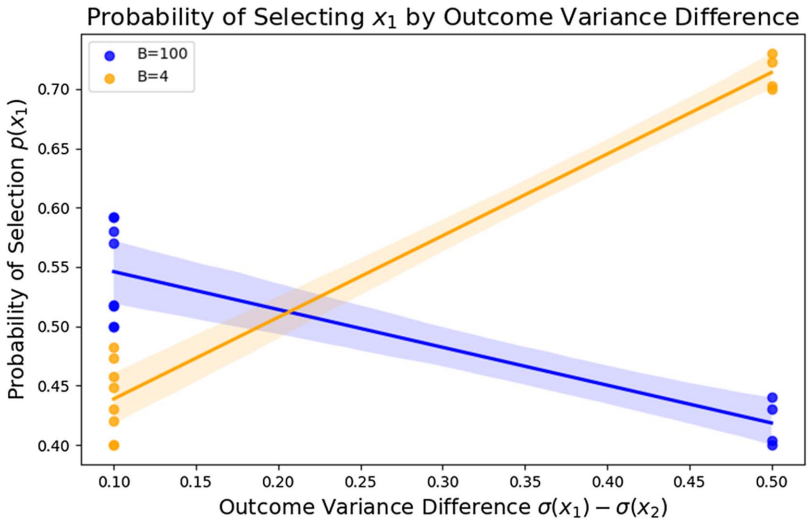
In this section, we describe two experiments conducted to evaluate the predictions of the proposed model of representation formation and behavior prediction. The two experiments differ in the behavior elicited from human participants. In Experiment 1, participants were presented with the probability and value of option outcomes, resulting in a decision-making setting; in Experiment 2, participants were not presented with option outcome values and probabilities directly and needed to learn through experience the relative outcome probabilities associated with different options.

In addition, in both experiments, participants were tested for their ability to detect changes in the stimuli using a standard visual change-detection task. Participants were not told beforehand when they would be tested for change detection, as initial stimuli did not signal which of the two tasks was being performed. These change-detection trials made up half of the trials in each episode, meaning the participants would need to remember information related to both the utility and the visual information of the stimuli (Figure 5).

Since participants do not know which of the two tasks they will be performing, they are required to form representations that are suitable for both tasks. First, it was predicted that on change-detection trials in Experiment 1, the difference between the utility of the two jars should impact the probability of identifying that a change in one of the jars had occurred. This is due to the assumption that representations are formed to balance both utility and visual

Figure 4

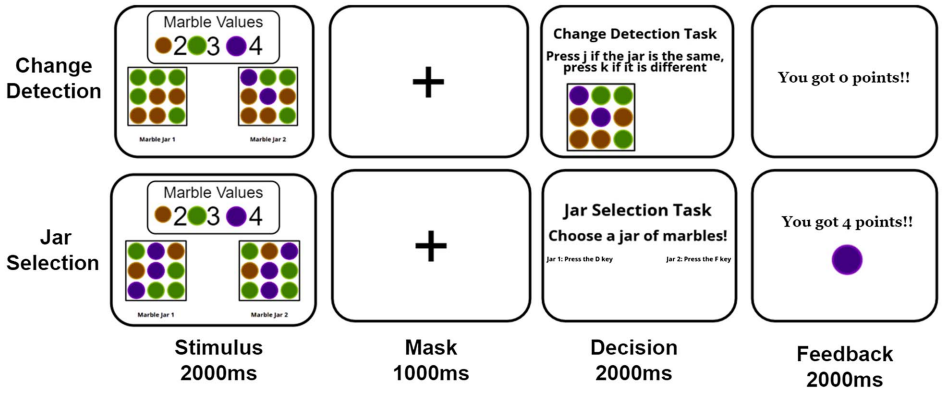
Probability of Selecting Marble x_1 Over x_2 by the Difference in the Outcome Variances of Marbles x_1 and x_2



Note. The lower β value demonstrates a risk-seeking preference by overvaluing higher variance option. The higher β value demonstrates a risk-averse preference by undervaluing the higher variance option. See the online article for the color version of this figure.

Figure 5

Experiment Paradigm Consisting of Change Detection Tasks (Top) and Jar Selection Utility-Judgment Tasks (Bottom)



Note. Times in milliseconds that each screen was viewed for are listed on the bottom. See the online article for the color version of this figure.

information, and that jars with a higher utility difference are thus easier to distinguish between.

Second, it was predicted that in Experiment 2, the utility difference of jars would initially be irrelevant to change-detection accuracy, since the utility associated with different outcomes is not known. However, as the participants learned the utilities of the marble jars through experience, the utility of marble jars should start to become relevant for predicting change detection. Finally, it was predicted that the jar selection of participants in Experiment 1 would consistently demonstrate risk-sensitive behavior, since they are told the specific value of each marble and the probability of receiving one, their behavior has the potential to be risk sensitive. Meanwhile, the risk-sensitive behavior in Experiment 2 was predicted to not initially occur as participants are unaware of marble utilities, but the sensitivity would increase over time.

Method

Three hundred participants from Amazon Mechanical Turk participated in the two experiments (150 participants in each experiment). Twenty-two participants were removed for failing to respond in time on too many trials, as defined by 3 *SD*s above the mean of all participants. All results are reported in relation to the remaining 278 participants (145 in Experiment 1, 133 in Experiment 2). For clarity, the first experiment is referred to as the decision-making experiment, since utilities are presented to participants, and the second experiment is referred to as the learning experiment, since the utilities are learned. Both experiments were approved by the Rensselaer Polytechnic Institute Internal Review Board. This study was not preregistered. Participant data, quantitative analysis, trained models, and analysis code are available on Open Science Framework (https://osf.io/h97s4).

Apparatus

On each trial of the experiment, participants were presented with two jars of nine differently colored marbles (orange, green, and purple). On trials of the decision-making experiment, participants were also presented with the values of these differently colored marbles. One of the marble colors was worth 2 points, another was worth 3 points, and the final was worth 4 points. The number of points associated with each color was counter balanced between participants. After a brief mask period, participants were told which of the two tasks they were to perform on that trial, either utility selection or change detection. Both experiments consisted of 10 blocks of 20 trials with 10 each of utility-selection and change-detection trials in each block.

On utility-selection trials, participants indicated using their keyboard if they would like to select the left (D) or right (F) jar of marbles to receive a randomly sampled marble from. In the decision-making setting, each marble color had an assigned point value. In the learning setting, the number of points associated with each marble color is randomized at the start of each block to between 2 and 4 points. To mask the values of these marble colors, a continuous distribution is created based on the marbles contained in each jar. This distribution is sampled from and rounded to 1 decimal place before being presented as the reward to the participant.

On change-detection trials, participants were presented with another jar of marbles on either the left or right side of the screen that was either the same or different from the jar of marbles they previously saw on the same side of the screen. Participants were then instructed to indicate by pressing a key on their keyboard whether the jar was the same (J) or different (K) from the jar they previously saw. In change-detection trials, participants received 3 points for correctly identifying the change, and in utility-selection trials, the participants received a range of approximately 2–4 points depending on the jar they selected.

Procedure

Before beginning the main experiment, participants received instructions and examples of the task and the two types of trials they would perform. Feedback was presented as a screen with a point score in the middle, appearing for 1 s. After receiving point feedback from making a decision, a progress bar at the top of the screen would fill in to indicate participants' accuracy in the current set of trials,

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filling the progress bar required 50, out of the approximately 70-point maximum per trial block. Participants were told they would receive a base payment of $2 with an additional $0.25 bonus on each block that they were able to fill a progress bar tracking the number of points they received.

Decision-Making Experiment Results

Risk Sensitivity

To measure risk-sensitive biases in participant decision making, we limit our analysis to Experiment 1 in which the utilities of both jars of marbles are known throughout the experiment. The left plot in Figure 6 shows the probability that a participant selects an option based on the difference in outcome variances between the chosen and unchosen marble jar. From these results, we can see a relatively high probability of selecting the option with a lower expected utility, indicating that participants are not only considering expected utility in their decision making. This effect is in line with previous research in cumulative prospect theory (CPT) (Schmidt & Zank, 2008), which can motivate the parameterization of risk sensitivity relative to the variance of outcomes using a coefficient that describes the impact of outcome variance on decision making (Harrison & Rutström, 2008).

The left plot in Figure 6 shows the probability of selecting one marble over the other depending on the difference between the outcome variance of the two options. We split these results between participants that demonstrated more risk-averse behavior and those that demonstrated more risk-seeking behavior. This demonstrates that some participants are more likely to choose the option with a lower outcome variance (risk averse), while others are more likely to choose the option with a higher outcome variance (risk seeking).

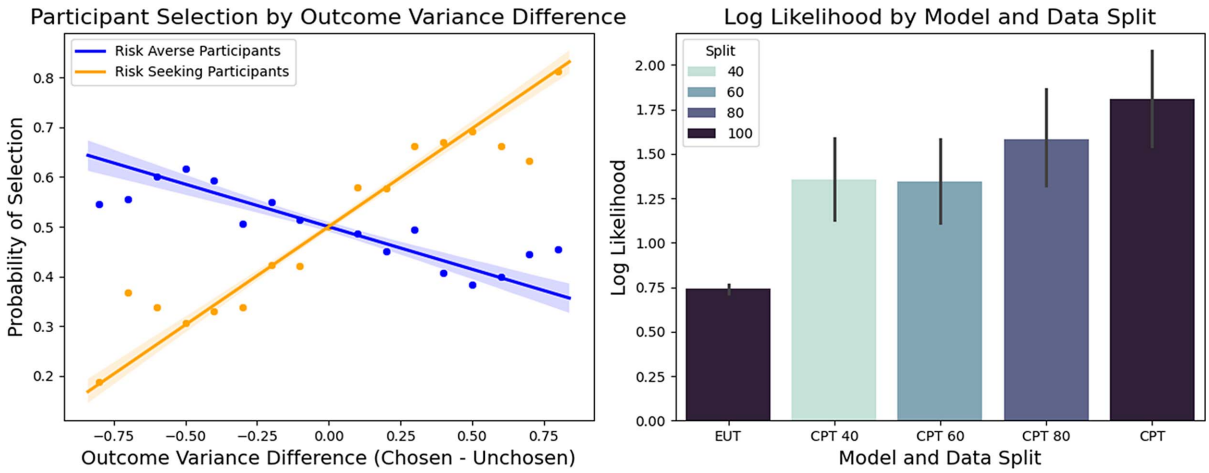
The right plot in Figure 6 shows the log-likelihood measure of predicting participant choice selection based on five different methods. A higher log-likelihood measure represents a closer accuracy to participant choice selection. The expected utility theory model predicts choice selection based solely on expected utility $E[x]$ of option outcomes. The CPT model predicts choice selection based on $E[x] - \lambda\sigma^2(x)$ where $\sigma^2(x)$ is the option outcome variance. This approach is inspired by prospect theory, which has frequently used this type of parameterization to predict risk-sensitive utility selection under uncertainty (Tversky & Kahneman, 1992). The value of λ was fit separately for each participant. For the results shown in Figure 6, we split the groups into risk-seeking and risk-averse groups of participants based on whether their fit coefficient was above or below 0, allowing us to compare the speed of learned risk-sensitive behavior relative to the individual participant's risk preferences.

To evaluate how well this λ parameter describes individual participant differences in general behavior, different fit and test splits were done to compare accuracy of predicting behavior that was not used to fit the parameter. These fit and test splits use a random percentage of an individual participant's decisions to fit the model parameter, and then use that fit to predict the left-out testing decisions. Fitting the λ parameter in this way demonstrates a negative impact on the likelihood of model predictions of participant behavior when a smaller set of decisions are used to fit parameters. This is represented by the bar color on the right plot of Figure 6, where the number indicates the percentage of participant decisions used to fit the parameter. A one-way analysis of variance (ANOVA) and Bayes factor (BF) t test (between 100 and 40 splits) compared the effect of the method of predicting participant behavior on the log-likelihood predictive accuracy ($F = 1.75$, $p = 4E\text{-}6$, BF10 = 1.099, power = 0.557). A post hoc Tukey test showed a significant difference between fitting the CPT method to all participant decisions and the 60% split ($p = .035$), and the 40% split ($p = .035$), but not significant effect for the 80% split ($p = .33$).

Previous accounts of risk-sensitive decision making have been used to accurately represent and predict decision making in this type of setting, it would typically be done by determining a

Figure 6

Participant Risk Sensitivity and Model Likelihoods



Note. Left: Participant probability of selecting an option based on the outcome variance difference between chosen and unchosen options, split between risk averse and risk seeking. Right: Log likelihood measure of predicting the participant's decisions based on EUT and CPT with different parameter fitting splits. Split indicates the percent of data points used to fit the parameter λ in the CPT model. Error bars represent 90% confidence intervals. EUT = expected utility theory; CPT = cumulative prospect theory. See the online article for the color version of this figure.

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risk-sensitivity coefficient for an individual participant, similar to what is shown in the left side of Figure 6. These differences in risk-sensitivity are often considered to be the result of differences in individual preferences (Harrison & Rutström, 2008), though results have indicated that personality types are similarly correlated with these behavioral differences (Filbeck et al., 2005) among other explanations. The model presented in this work offers an alternative explanation of the source of risk sensitivity, as resulting from the formation of information constrained representations of the marble jars in the task. This explanation is supported by a later analysis of the representations formed by the proposed model while it is predicting the behavior of individual participants based on the stimuli they observe.

Change Detection Bias

We next investigated the probability of detecting a change between the initial and memory probe stimuli, based on the difference in the expected utility between the two stimuli. This analysis is limited to the case where a change did occur, as it allows us to compare the relationship between change-detection accuracy and expected utility difference. These results are shown in Figure 7 and show a positive relationship between the difference of the stimuli utilities, and the probability that a participant would detect that a change has occurred (Pearson correlation $R = 0.87$, $p = 1E\text{-}5$). In the middle plot of Figure 7, we compare this with the probability of detecting change based on visual information, as measured by the percent of marbles that have changes, this demonstrates as slight positive correlation, but it is insignificant (Pearson correlation $R = 0.39$, $p = .30$).

On the right plot of Figure 7, we compare the log likelihood of predicting participant change detection based on different approaches. The visual method is based on the percentage of marbles that have changed between the original S_1 and memory probe S_2, called $D(S_1, S_2)$. The four utility methods use a weighted value of the same visual difference and the mean-squared utility difference between the original and new jar of marbles $D(S_1, S_2) + \kappa(V(S_1) - V(S_2))^2$ where $V(S)$ is the mean utility of a stimulus, and κ is a parameter that is fit to participant behavior using different splits of fitting and testing, as was previously done in the risk-sensitivity case. While it might be assumed that stimuli that have a higher utility difference are more visually different, an analysis demonstrated no clear correlation between utility and visual similarity, this is shown in the Figure A1.

These results show the same effect as with predicting participant risk sensitivity using a fit parameter. When the whole set of change-detection decisions is used, there is a high likelihood of the model, but when fitting this κ parameter to a limited random sampling of participant change-detection decisions and using it to predict that participants' other decisions, the likelihood is significantly decreased. A one-way ANOVA and Bayes factor t test (between 100 and 40 splits) compared the effect of the method of predicting participant behavior on the log-likelihood predictive accuracy ($F = 9.24$, $p = 2.76E\text{-}7$, BF10 = 1.016, power = 0.543). A post hoc Tukey test showed a significant difference between fitting the utility method to all participant decisions and the 80% split ($p = .025$), the 60% split ($p = .017$), and the 40% split ($p = .017$).

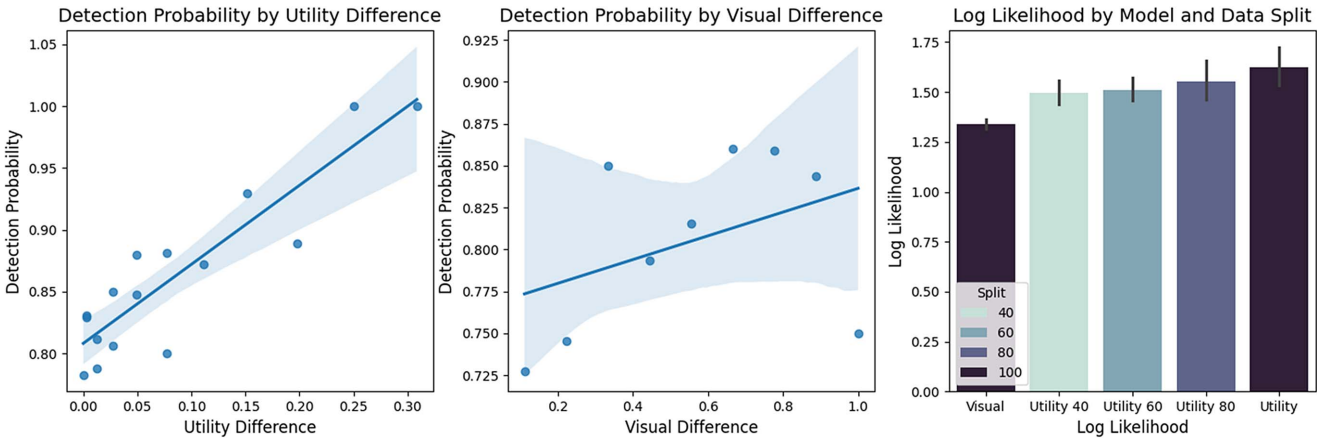
Learning Experiment Results

Learned Risk Sensitivity

Predicting utility selection in the learning experiments requires a method that takes into account the fact that participants do not know the utility outcome of options early on in trials and must learn them through experience. As a result, participants are not expected to demonstrate a risk-sensitive bias on earlier trials, whereas on later trials this effect should be observable. This is confirmed in the left side of Figure 8 by comparing the slope of the regression of option outcome variance and the probability of selecting that option, as the learning experiment progresses. This figure shows that the

Figure 7

Participant Change Detection Bias and Model Likelihoods

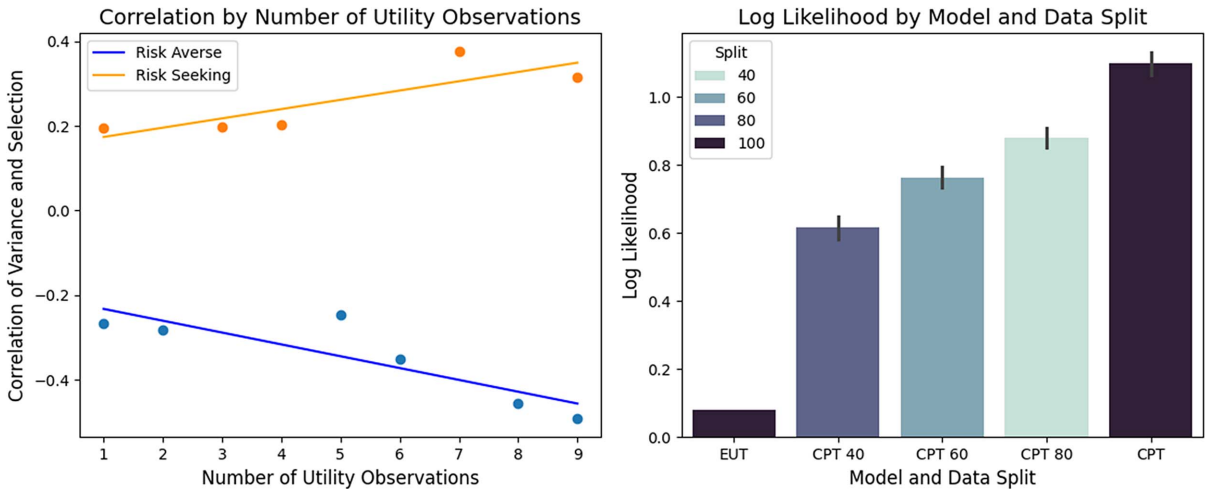


Note. Left: Participant probability of detecting a change between two stimuli based on the difference in utility between those stimuli, as measured by mean-squared error. Middle: Participant probability of detecting a change between two stimuli based on the visual difference between those two stimuli, as measured by the percent of marbles that have changed. Right: Log likelihood of the visual based predictions of participant behavior compared to the utility weighted predictions using four different splits of fitting the utility parameter u. See the online article for the color version of this figure.

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Figure 8

Participant Learned Risk Sensitivity and Model Likelihoods



Note. Right: Log likelihood of expected utility theory and cumulative prospect theory methods of predicting participant selections. Split indicates the percent of the individual participants decisions that were used to fit the learning-rate α and risk-sensitivity λ parameters. EUT = expected utility theory; CPT = cumulative prospect theory. See the online article for the color version of this figure.

probability of detecting a change that has occurred is positively correlated with the utility difference between the two stimuli, and that this relationship is stronger than the correlation between change detection and visual difference.

This would be impossible with the standard CPT inspired model that was used in the first experiment. To account for this updating of the effect of risk sensitivity, we compare a model that uses an additional α parameter to adjust the effect of the λ risk-sensitivity parameter as the participant observes more utility-selection outcomes. This is done according to the equation $\lambda_{i+1} = \lambda_i + (1 - \alpha)(\lambda_i - \lambda)$ where λ is the final coefficient, and λ_i is the coefficient used for the *i*th utility selection. As with the previous comparison in Figure 6, we again split the groups into risk-seeking and risk-averse groups in Figure 8. This shows that risk-seeking and risk-averse participants make decisions closer to expected utility maximization after one utility observation, but that their behavior diverges as the participants learn the utilities associated with option features. Only correlations with a *p* value of under .05 are included in this comparison, resulting in 12 of 18 correlations present in the left side of Figure 8.

The results on the right side of Figure 8 demonstrate the increased sensitivity that the CPT method has on learning data compared to decision-making data. This is likely due to the combined sensitivity introduced by the additional parameter α which is required to reflect the updating risk sensitivity that is shown on the left side of Figure 8. A one-way ANOVA and Bayes factor *t* test (between 100 and 40 splits) compared the effect of the model type on log-likelihood predictive accuracy of participant performance ($F = 456.93$, $p < 1E\text{-}10$, BF10 = 3.277E39, power = 1.0). A post hoc Tukey test demonstrated a significant difference in log likelihood between all method comparison pairs ($p < 1E\text{-}6$ for all). The model presented in this work seeks to represent the updating risk sensitivity of learned behavior without requiring fitting coefficients to participant data in this highly sensitive way.

Acquired Equivalence

Finally, we examined the previously discussed change-detection bias shown on the left side of Figure 7, but in the learning setting of the second experiment. If participants are forming visual representations that balance the two tasks at hand, and take into account their cognitive constraints, then this bias should not appear on earlier trials of the task when utility is not known to a high degree of accuracy. When utility has been learned through experience, the previously observed bias that occurred in the decision-making experiment is expected to impact behavior in the learning task.

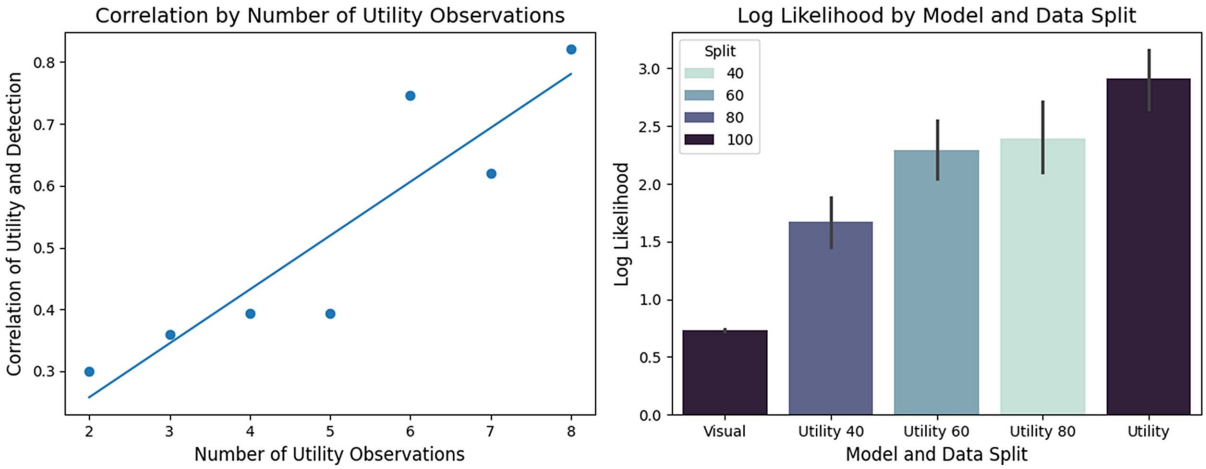
This effect is demonstrated in Figure 9 by performing the same regression analysis done in Figure 4, but only performing the linear regression on the subset of change-detection judgments made after observing a specific number of utility observations, indicated by the *x*-axis. This allows us to compare the impact that learning the utility of marble jars has on the potential bias of utility in the change detection. As shown on the left side in Figure 9, early trials of the task show low correlation with stimuli utility difference and change-detection accuracy. However, on later trials of the task, this positive correlation appears (Pearson correlation $R = 0.91$, $p < .005$). One explanation for this difference in change-detection accuracy is that as utility is learned participants attend to different areas of the stimuli, resulting in a change in visual representation. Another explanation for this updating representation similarity is that the visual information that is attended to is the same, and that there is a change in how information is translated from working memory to more long-term visual representations used for later recall.

A one-way ANOVA and Bayes factor *t* test (between 100 and 40 splits) compared the effect of model data split on log-likelihood accuracy in predicting participant behavior ($F = 45.45$, $p < 1E\text{-}6$, BF10 = 2.765E9, power = 1.0). A post hoc Tukey test showed a significant difference between each model type ($p < 1E\text{-}6$), except for utility and utility 80 ($p = .5$). These results demonstrate an

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Figure 9

Participant Acquired Equivalence and Model Likelihoods



Note. Left: Slope of regression fit to change-detection bias by utility, as the number of utility observations increases. Right: Comparison log likelihood of predicting participant behavior between visual similarity and utility weighted visual similarity. Split indicates the percent of the data that was used to fit the learning-rate α and utility weight parameter κ. See the online article for the color version of this figure.

acquired equivalence of marble jars based on the utility associated with them. On early trials, there is no correlation between the ability of participants to detect that a change has occurred and the utility of the original and changed jar of marbles. However, as the utility of the jars of marbles is learned through experience, they begin to demonstrate the same bias in change-detection ability based on utility that was observed in the first experiment.

Model Results

In all modeling results, the model observes the utilities actually presented to the participants in the decision-making setting, and the reward feedback shown in the learning setting. The utility of the change-detection task is not considered directly, outside of the fact that participants who paid more attention to this task would perform better, resulting in more accurate change detection, which would change the parameters of the model predicting behavior.

Risk Sensitivity

The results in Figure 10 demonstrate the same risk-sensitive effect that is observed in participants while they are engaged in the task. When the value of the β parameter is low and the information bottleneck on the model representations is small, the behavior is risk averse, preferring lower outcome variance options. When the value of the β parameter is high, and the information bottleneck is tighter, the behavior becomes risk seeking. This risk-sensitive behavior of the model is a result of its training to balance utility accuracy and reconstruction accuracy, in relation to the information bottleneck imposed by the model. The variance in risk sensitivity that results from changing this β parameter allows us to fit individual participant behavior to a β parameter and reflect their individual differences in behavior.

This shows that the proposed model can produce humanlike biases in the proposed experiment while taking in as input the actual visual stimuli shown to participants, instead of handcrafted features or descriptions of choice outcomes that other models rely on. In addition, the source of these biases is the information constraint and training method of the proposed model, whereas alternative explanations of the source of these biases rely on additional parameters to explain biases in human behavior.

Comparing the model log likelihood by data split shows that fitting the β parameter to a limited set of observations from participants does not have the same negative impact on predictive accuracy as with the CPT inspired model shown previously. A one-way ANOVA and Bayes factor t test (between 100 and 40 splits) compared the models by log-likelihood accuracy and found no significant difference between parameter fit and accuracy ($F = 1.55$, $p = .199$, BF10 = 0.813, power = 0.500). This is a key benefit of the proposed model, as it can make accurate predictions of participant behavior from a limited set of their decision making.

Utility-Biased Change Detection

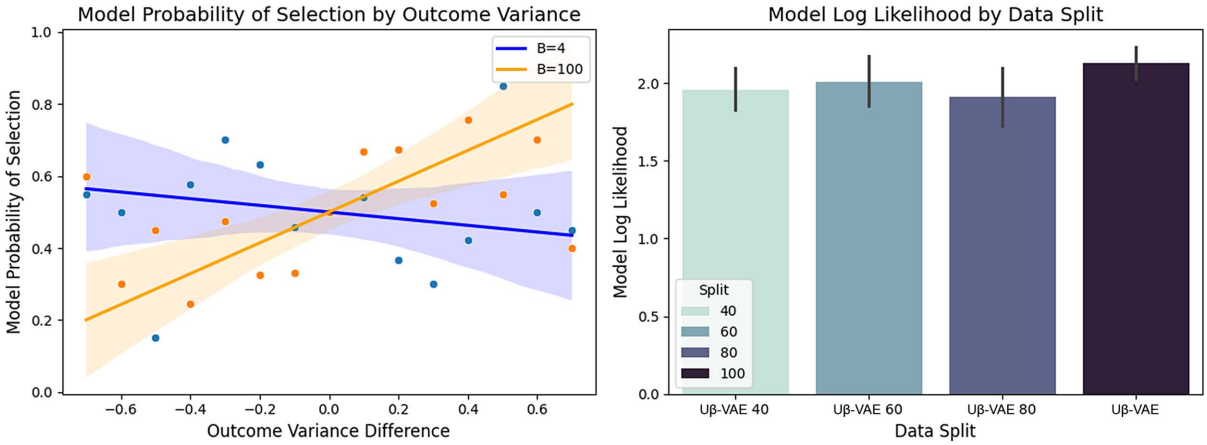
We next investigated the change-detection bias based on the difference in utility of the original and changed stimuli. This is an important effect observed in participants, because it supports the hypothesis that representations of visual information are formed to balance reconstruction accuracy, utility accuracy, and done so in relation to cognitive constraints. If visual representations could be formed in a way that retains utility information and visual information using separate mechanisms, then this bias that negatively affects performance in the task would not be present. In addition, if utility information was represented in a way that was significantly less complex than visual information, again there would not be the same behavioral bias. Thus, these results support the hypothesized balance that motivates the structure and training of our proposed model.

To predict change detection, the Uβ-VAE model compares the similarity between the original and changed stimulus representations z, as measured by mean-squared difference. As a result, the

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Figure 10

Uβ-VAE Model Risk Sensitivity and Model Likelihoods



Note. Left: Uβ-VAE model probability of selecting a stimulus by the outcome variance difference between that stimulus and the alternative stimulus, demonstrating the risk-sensitive effect based on the value of β. Right: Log likelihood of model predictions based on the split of the participant data that was used to fit the β and υ parameters. Uβ-VAE = utility β-variational autoencoder. See the online article for the color version of this figure.

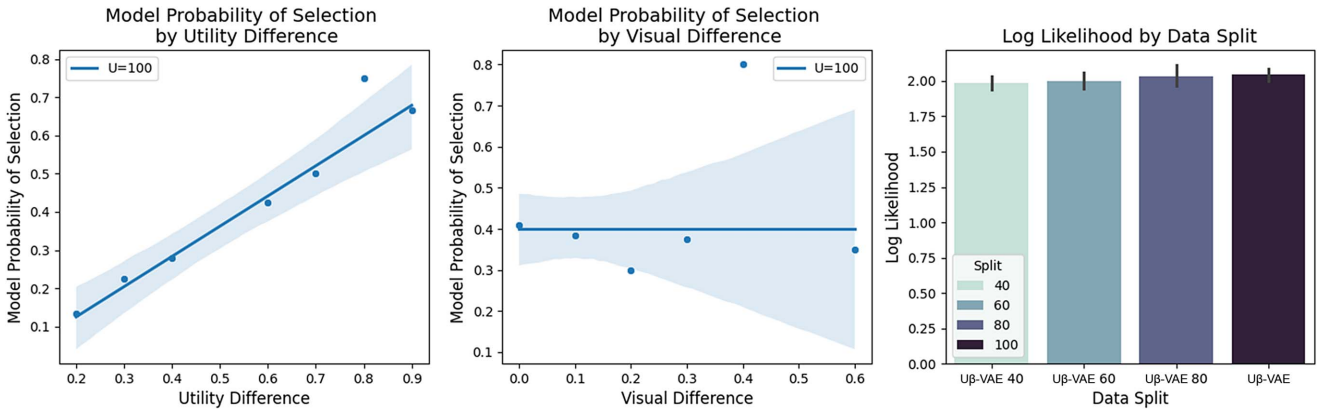
model does not need to directly fit individual performance using a separate parameter based on the influence of utility. Instead, it fits to the β and υ parameters which define the information constraint and the importance of utility prediction accuracy.

Results shown in Figure 11 demonstrate that using a high value of the utility parameter υ produces a strong correlation between utility difference and the probability of detecting a change. Comparatively there is no clear correlation between visual difference and model probability of detecting a change. This is a result of the high information bottleneck and high weighting of the accuracy of utility predictions, which means representations are mostly concerned with accurately reflecting utility differences. This provides further support of the hypothesis regarding representation formation, as the Uβ-VAE model is explicitly trained to balance reconstruction and utility prediction accuracy in relation to a constraint. Thus, the proposed model suggests that the observed utility difference bias in change-detection accuracy from human participants stems from the use of the same cognitive resources to represent utility and visual information.

A one-way ANOVA and Bayes factor t test (between 100 and 40 splits) compared the models by log-likelihood accuracy and showed no significant difference between parameter fit split size and accuracy ($F = 0.89$, $p = .44$, BF10 = 0.256, power = 0.222). While the result showing the probability of the model detecting a change based on visual information alone may seem surprising, it is

Figure 11

Uβ-VAE Model Change Detection Bias and Model Likelihoods



Note. Left: Uβ-VAE model probability of detecting a change in stimuli by the difference in the utility between in initial and changed stimulus. Middle: Uβ-VAE model probability of detecting a change in stimuli by the difference in visual information, as measured by the percent of marbles that are different. Right: Log likelihood of model fit to parameters using different splits of individual participant change-detection decisions. Uβ-VAE = utility β-variational autoencoder. See the online article for the color version of this figure.

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important to remember that the model does not predict change detection based on visual information, but rather the similarity of the original and changes stimuli latent representations z which contains information related to both visual similarity and outcome utility.

Learned Risk Sensitivity

The same effect of learned risk sensitivity that was observed in participants can be tested with our proposed model by taking the correlation of outcome variance and the probability that the model will select an option, and observing this correlation as the model learns the utilities associated with options. This is done using different values of the β parameter which previously were shown to result in different risk preferences in utility selection in the Uβ-VAE model, shown in Figure 12, representing the difference in risk sensitivity between participants.

A one-way ANOVA and Bayes factor t test (between 100 and 40 splits) compared the models by log-likelihood accuracy and showed no significant difference between parameter fit split size and accuracy ($F = 1.55$, $p = .19$, BF10 = 0.305, power = 0.267). Thus, the proposed model is as accurate in predicting behavior when using a smaller split of data from human participant decisions as it is when using all of the decisions they made. This means that our proposed model is able to accurately predict the speed of learned risk sensitivity in participants as a result of their individual differences in information processing capacities. This also ties in the individual risk-sensitivity differences to acquired equivalence, as they both result from using models fit to different information processing capacities.

Acquired Equivalence

In the decision-making setting (Figure 7), biased change detection was observed in all trials because the participants were explicitly presented with marble utilities. In the learning setting, we would expect the same plot of change detection and utility difference to show no bias early on in learning, and a more significant bias later in learning. The expected effect can be observed in Figure 9 which plots the utility-detection ratio described in Figure 13 as a function of the number of utility judgments that have previously been made in the block. This utility-detection ratio increases as the participants make more utility judgments, and their change detection becomes more biased by the utility of the stimulus.

The right-hand side of Figure 13 shows the likelihood metric using the same splits of model fitting and testing previously described. Compared to the previous model that predicted change detection using the visual similarity and utility similarity of the initial and changed stimuli, the proposed model demonstrates little difference in likelihood when trained on only a subset of observations. A one-way ANOVA and Bayes factor t test (between 100 and 40 splits) compared the models by likelihood and demonstrated no clear effect of the model parameter fitting data split ($F = 0.47$, $p = .69$, BF10 = 0.297, power = 0.260).

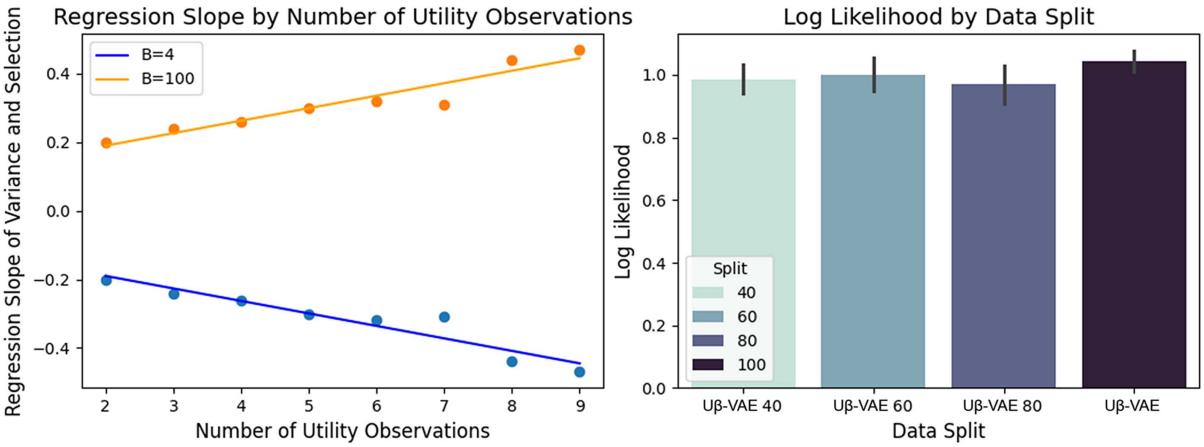
Model Comparisons

We next performed an analysis across models with different numbers of parameters on the model predictive accuracy mean and standard deviation, and the Bayesian information criterion. This metric was chosen due to the difference in the number of parameters between the models under comparison, to provide support that our proposed model is not merely relying on more parameters to increase predictive accuracy. Predictive accuracy refers to the average probability that the model assigned to the choice that the participant made. In the following tables, bold values indicate statistical significance of higher predictive accuracies compared to other models in the same data split for that type of trial.

In the first experiment on decision making and change detection, the accuracy of the proposed model significantly outperforms the CPT inspired model on all splits of parameter fitting and testing

Figure 12

Uβ-VAE Model Learned Risk Sensitivity and Model Likelihoods

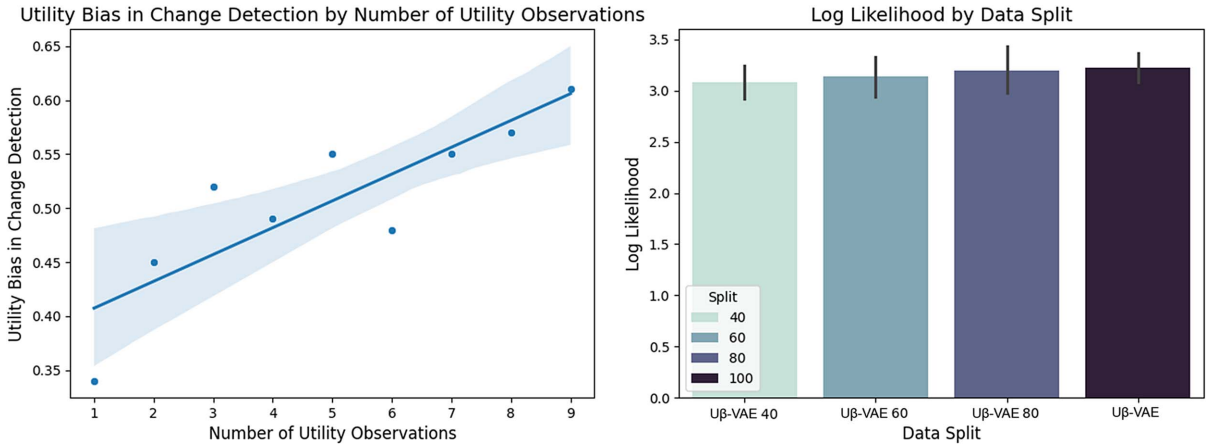


Note. Left: Slope of regression fits to the Uβ-VAE model probability of selecting an option by its variance comparing the low and high values of the β parameter, this compares both risk-averse and risk-sensitive effects. This is done with the full data for fitting parameters. Right: Log likelihood scores using different data splits to fit the model parameters. Uβ-VAE = utility β-variational autoencoder. See the online article for the color version of this figure.

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Figure 13

Uβ-VAE Model Acquired Equivalence and Model Likelihoods



Note. Left: Slope of regression fit to the Uβ-VAE model probability of detecting change by stimuli utility difference, as the number of utility observations increases. Right: Log likelihood scores by data split. Uβ-VAE = utility β-variational autoencoder. See the online article for the color version of this figure.

in utility-selection trials. In the change-detection trials, our proposed model significantly outperformed all data splits that used less than 100% of the full training data to fit individual participants' model parameters (Table 1).

Decision-Making Experiment 1

In the second experiment on utility learning and change detection, the accuracy of the proposed model significantly outperforms utility judgment task accuracy in predicting participant behavior of the CPT model on all data splits for fitting parameters. In the change-detection task, the proposed model outperforms the visual based prediction model on all fittings that use less than 80% data split. Comparing the second experiment in learning shows a larger difference in the accuracy of all models in predicting utility selection compared to change detection, due to the fact that utility was not presented to participants and instead learned through feedback. In addition, this difference likely meant that participants were able to allocate more attention to change detection, especially on earlier trials, leading to higher change-detection accuracy which was easier to predict for all models (Table 2).

Learning Experiment 2

Overall, these fitting results demonstrate that models like the CPT and visual utility-based prediction approaches can accurately predict human behavior when using all data available from that participant. However, the proposed model is unique in that it does not parameterize the observed behavior and account for it through data fitting, but rather explains it on the basis of an information

Table 1

Table of Predictive Accuracies in Mean and Standard Deviation and BIC Metrics for Each Split of Model Comparison for the First Experiment on Decision Making

Model	Parameter	Trial type	100% split Accuracy (BIC)	80% split Accuracy (BIC)	60% split Accuracy (BIC)	40% split Accuracy (BIC)
EUT	None	Utility selection	52.23 ± 0.03 (N/A)	N/A	N/A	N/A
CPT	λ	Utility selection	79.4% ± 3.80 (59.52)	79.4% ± 3.88 (73.9)	74.1% ± 2.59 (93.7)	73.9% ± 2.79 (92.88)
Uβ-VAE	β, υ	Utility selection	**85.6% ± 4.02 (32.14)**	**85.5% ± 13.0 (35.28)**	**85.0% ± 8.90 (33.87)**	**85.8% ± 7.20 (34.56)**
Visual	None	Change detection	73.7% ± 0.03 (N/A)	N/A	N/A	N/A
Utility	κ	Change detection	80.25% ± 4.32 (80.55)	78.81% ± 4.04 (79.24)	77.9% ± 1.69 (76.03)	77.5% ± 1.90 (71.02)
Uβ-VAE	β, υ	Change detection	80.4% ± 0.82 (57.18)	**80.3% ± 0.92 (57.5)**	**80.0% ± 1.41 (58.4)**	**80.3% ± 2.07 (59.29)**

Note. Bolding represents significantly better models ($p < .05$). BIC = Bayesian information criterion; EUT = expected utility theory; NA = not applicable; CPT = cumulative prospect theory; Uβ-VAE = utility β-variational autoencoder.

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Table 2

Table of Predictive Accuracies in Mean and Standard Deviation and BIC Metrics for Each Split of Model Comparison for the First Experiment on Learning

Model	Parameter	Trial type	100% split Accuracy (BIC)	80% split Accuracy (BIC)	60% split Accuracy (BIC)	40% split Accuracy (BIC)
EUT	None	Utility selection	51.96 ± 0.00 (N/A)	N/A	N/A	N/A
CPT	α, λ	Utility selection	62.7% ± 0.04 (43.50)	54.7% ± 0.03 (79.86)	49.7% ± 0.04 (105.42)	42.5% ± 0.04 (146.91)
Uβ-VAE	β, υ	Utility selection	**65.89% ± 0.82 (31.30)**	**65.5% ± 2.0 (32.11)**	**64.8% ± 1.4 (34.97)**	**64.2% ± 1.0 (37.44)**
Visual	None	Change detection	70.7% ± 2.30 (N/A)	N/A	N/A	N/A
Utility	α, κ	Change detection	**94.5% ± 2.54 (34.38)**	90.8% ± 3.31 (44.9)	89.8% ± 2.61 (47.93)	81.1% ± 1.66 (75.19)
Uβ-VAE	β, υ	Change detection	91.6% ± 4.02 (31.44)	91.0% ± 8.95 (33.75)	**91.8% ± 7.25 (33.09)**	**91.5 ± 13.0 (34.44)**

Note. Bolding represents significantly better models ($p < .05$). BIC = Bayesian information criterion; EUT = expected utility theory; NA = not applicable; CPT = cumulative prospect theory; Uβ-VAE = utility β-variational autoencoder.

constraint (β) and utility attention weight (υ). The result is a model that is able to predict unobserved data from participants to a higher degree of accuracy compared to alternative approaches.

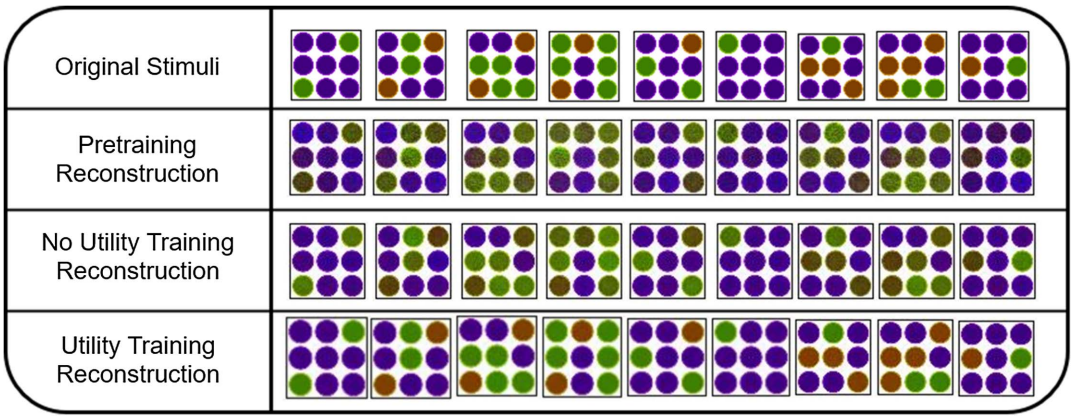
An important aspect of the train-test splitting of our proposed model is that it does not use different fitting for the two types of trials, as opposed to the other models used for comparison. When predicting behavior in both types of trials, the β and υ parameters are fit using decisions made in both types of trials and applied onto the remaining data. While this train and testing split is more limited in the directness of testing predictive accuracy compared to a method like Bayes factors analysis, there is added difficulty in evaluating cognitive models that incorporate ad hoc trained generative neural networks, due to long training times required for each run of evaluation. As a result, the current statistical analysis should be viewed as providing support for a practical difference between our proposed model and alternatives.

Source of Model Behavior

To better understand the source of the observed model biases in change-detection accuracy and utility selection, we performed an analysis of the reconstruction of marble jars with and without utility-based training. This was achieved by training for an initial pretraining period of 100 epochs with the utility parameter υ set to zero, and then training for an additional 100 epochs with and without a positive utility parameter. The results shown in Figure 14 demonstrate the impact that utility-based training has on reconstruction accuracy, as including this parameter allows the model to accurately reconstruct the high value orange marbles. However, training without a high utility parameter produces biases in the reconstruction which favor the most common purple marbles. This effect is in line with expectations of an information-constrained system, which should allocate informational resources onto areas of the visual scene that

Figure 14

Uβ-VAE Model Stimuli Reconstructions



Note. Comparison of Uβ-VAE model reconstructions of a subset of experiment stimuli based on an initial pretraining without utility learning, a continued training with no utility learning, and a continued training with utility information. Uβ-VAE = utility β-variational autoencoder. See the online article for the color version of this figure.

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have the biggest improvement on reconstruction accuracy, which are the more common marbles. Thus, training with a positive utility weight parameter can produce more accurate reconstructions when the information bottleneck is sufficiently tight that training without it results in only the most common marble in the jar being reconstructed faithfully.

The relationship between this bias in reconstruction accuracy and the model's bias in change-detection accuracy based on utility can also be connected to the risk-averse bias in utility selection. This can be shown experimentally through simulation by introducing noise into the evaluation of a marble jar's utility. Adding noise allows us to compare the standard deviation of utility outcomes to the average distortion in reconstructing the marble jar stimuli. This noise takes the form of a 10% probability that a higher value marble will be mistakenly identified as lower value. Comparing the results shown in Figure 15 shows the predicted utility of the entire set of marble jars with this noisy utility evaluation, showing that the bias in predicted utility matches the effect of risk-averse behavior. This is achieved by setting the information constraint parameter in such a way to produce the same risk-averse behavior previously observed. This is closely related to the ideas of CPT which predict the over attention to outcomes based on their probability or value, however, it explains the source of these attentional biases as resulting from a balanced optimization of representation accuracy and utility, in an information-constrained model.

The results shown in Figure 16 demonstrate that in early trials, before utility is learned, there is no clear bias in the similarity of the marble jar stimuli based on utility. This conforms to what is expected from the stimuli set, since there is generally no visual information similarity and utility similarity. Since the model has not yet learned to form representations that additionally encode stimuli utility, they are not yet biased based on stimuli utility. This also provides evidence that there is no systematic bias in the data set of stimuli that make marble jars with similar visual information generally more similar in expected utility.

On later trials of the task, once utility has been learned, there is a clear bias in how the stimuli information is being represented based on the utility of the stimuli. This explains the bias in change-detection accuracy that is present in our model, as well as the high predictive accuracy in determining whether the participants in the task will recognize a change in the stimuli. This is a key result related to the broad hypothesis that visual representations are formed to balance utility and accuracy, in a manner related to individual information processing limitations.

Discussion

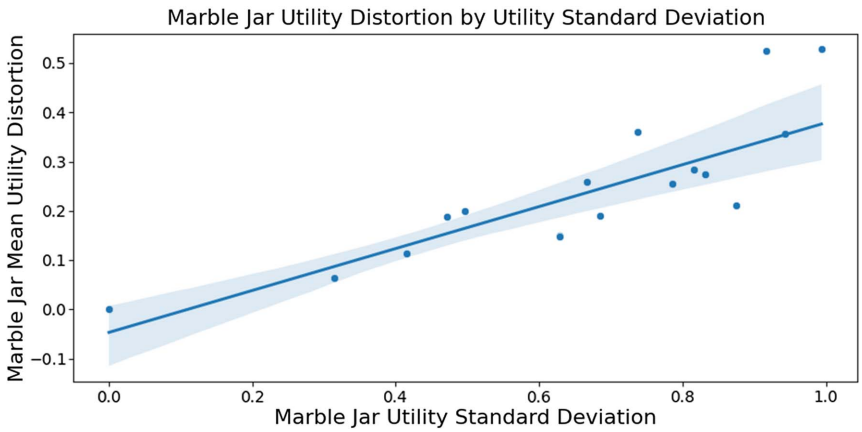
Risk sensitivity and acquired equivalence are both well studied phenomena within the decision science literature. However, the specific source of these effects and how they relate to individual differences remains poorly understood. This understanding can be particularly difficult to achieve, as predicting these effects as resulting from a mathematically formulated bias that uses coefficients fit to individual behavior demonstrates poor performance in generalizing to other behavior observed by that participant, let alone different tasks. This was demonstrated in the experiments presented here by randomly splitting participant behavior into fitting and testing groups, which showed the poor accuracy in predicting individuals' behavior based on data that was not observed in fitting parameters.

The model in this work seeks to further the understanding of the source of these effects by predicting utility-based decision making and visual change detection in a single experiment paradigm. Under this view, individuals' level of risk sensitivity and the speed and degree of their acquired equivalence is in part a result of their individual differences in information processing capacity. The benefit of this understanding is that it produces predictions that are more closely related to immutable differences in individuals, rather than the view of these differences as merely preferential. A comparison of the accuracy of this model in predicting participant behavior while splitting their decisions into separate groups demonstrated the benefits of conceptualizing risk sensitivity and acquired equivalence in this way.

While there are many positive aspects of this model, particularly in its ability to accurately predict performance using a limited number of data observations, there are limitations that point to future

Figure 15

Simulation of Noise Introduced Into Marble Jar Information Representing a Constraint Applied to Visual Information Representations, Which Produces a Risk-Sensitive Effect

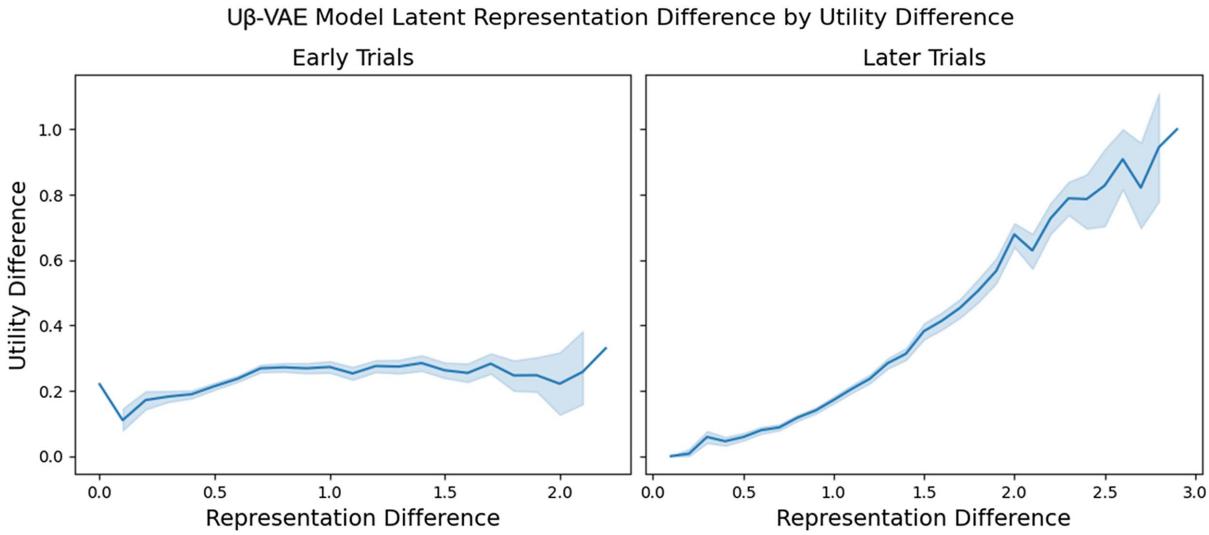


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Figure 16

Utility Difference of All Pairs of Stimuli by the Difference in Their Representations, as Measured by MSE



Note. Left: Early trials of the task before utility are learned. Right: Later trials of the task after utility are learned. MSE = mean squared error; Uβ-VAE = utility β-variational autoencoder. See the online article for the color version of this figure.

directions of expanding this approach. One of the most notable limitations of this model is that it is limited to producing memory representations of visual information and utility predictions that are useful for simple judgment tasks. Recent methods in generative artificial intelligence, of which the β-VAE is an example, focus on the representation and generation of natural language (Brown et al., 2020), audio (Kim et al., 2019), motor control (Ren & Ben-Tzvi, 2020), as well as combinations of these through multimodal approaches (OpenAI et al., 2023). Another limitation to the model is the lack of further statistical analysis that is prevented due in part to it relying on ad hoc trained generative models, which makes more complex statistical analysis difficult. One way to address this in the future could be through the use of pretrained models, such as those used in (Bates et al., 2024).

The proposed model is an instance of a general connection between generative artificial intelligence and cognitive modeling approaches, which could conceivably be applied to modeling learning and decision making in more broad domains. Several methods have recently been developed that sought to apply these approaches onto cognitive modeling in these domains, such as the use of generative models to predict phishing email detection (Xu et al., 2022), motor control (Taniguchi et al., 2022), auditory learning (Beguš, 2020), and multimodal learning (Ivanovic et al., 2018). We view our proposed model as an instance of a larger family of models that connect generative artificial intelligence in cognitive modeling, which has many applications for future study.

Utility Selection Biases

Deviations from behavior that maximizes expected utility are perhaps one of the most well studied phenomena in decision science. To date, many models have sought to explain human deviations from expected utility maximization, which can be done in a variety of methods that have demonstrated connections to human psychology, neuroscience, and computer science. Often these models use parameters that are fit to human behavior and express deviations from selecting options with higher expected utility by altering their true expected utility through these parameters. However, these cognitive models are fundamentally not attempting to choose options that maximize utility, but rather reflect observations of human decision making.

The model proposed in this work takes an alternative approach by consistently attempting to choose the option with the highest expected utility, but doing so in an information-constrained manner in a method that relies on the visual information presented to a participant. In this sense, the error of the utility predictions of the proposed model are not merely the result of fitting parameters to maximize the likelihood of all observed behavior. Rather, the model parameters are optimized to express the information processing capacity and attentional biases in participants, which should be a more generalizable aspect of behavior. To support this, we performed splits of participant behavior and used a limited number of observations to optimize these model parameters. Results from this comparison demonstrated that the proposed model can be fit to a limited sampling of human utility-selection decisions and still predict unobserved behavior at a higher accuracy compared to alternative methods.

Change Detection Biases

Memory and decision making are highly interdependent and performing two tasks simultaneously, such as in the experiments introduced here, requires efficiently allocating limited informational resources to achieve multiple behavioral goals. One way that decision making can impact memory is through acquired equivalence, which causes stimuli that are associated similarly in outcomes (Hall et al., 1993) to be represented more similarly (Meeter et al., 2009). Our experiment required participants to keep the utility and

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visual information associated with stimuli in mind before they were shown which task they were to perform. The result of this was that in the first experiment the ability of participants to detect a change between two stimuli was immediately influenced by their difference in utility. In the second experiment, this effect was not immediately observed but did become present once the utility of stimuli was better known. This effect could be modeled in the experiments presented here by adjusting the visual similarity of stimuli pairs based on their expected utility. However, this approach is again not attempting to detect change as accurately as possible, but rather attempting to account for an observed behavioral bias in humans.

The model we propose takes inspiration from recent neuroscientific findings in computational models of biological perception, with the addition of a bias based on stimuli utility. The result is a model that attempts to accurately detect the change observed by participants but does so by forming low-dimensional representations of these stimuli that are also used to predict their utility. Because our proposed model balances visual similarity, utility predictions, and an information constraint, it is unable to perfectly detect these changes. These competing goals result in predictions of change detection that accurately conform to human behavior and explain the mechanics behind acquired equivalence as resulting from prolonged experience interacting with stimuli in a task that requires predicting their utility.

Modeling Results

The model comparisons we present demonstrate the effectiveness of our model in predicting biases in both change detection and utility selection, while also allowing for improved accuracy in predicting participant behavior that was not used to fit model parameters. This is a highly relevant comparison for models that seek to not only explain human behavior but replicate it in a manner that is biologically plausible. Since alternative approaches focus on explaining biases rather than performing tasks as well as possible, they have poor generalization when predicting behavior that was not previously observed when fitting parameters.

Overall, these modeling results demonstrate a clear connection between the biases present in human participant behavior. A key feature of the proposed model is that it is trained to balance utility accuracy, reconstruction accuracy, and informational complexity in a way described by the VAE model. The similarity of model and human behavior supports the previously discussed hypotheses regarding the nature of visual representation formation in humans, as alternative explanations that do not consist of the same balance of task relevance and visual information, mediated by informational complexity, would not produce the same behavioral biases.

Conclusions

The model presented in this work seeks to explain two commonly observed phenomena in human decision making, risk-sensitive decision making and task-relevant attention. The proposed model achieves this through an explanation of these biases as resulting from the formation of information-constrained representations of visual information. The specific task used in these experiments was based on making utility judgments, which had the benefit of connecting representation formation strategies to risk-sensitive behavior. However, the implications of these results are not only in relation to utility judgment biases but are generally applicable to understanding how humans form representations of visual information. These phenomena are closely related to broad cognitive faculties such as attention and statistical judgments. These faculties have long been understood to be biased in humans, though traditionally this was seen as the result of heuristics, or explained by cognitive limitations without a formal explanation for how these limitations produce biased behavior.

Experimentation results comparing model behavior to human participants in the two experiments provide support for an explanation of the source of biases in human decision making from attention and statistical judgments. Their cause is explained by the formation of representations of visual information that allocate more informational resources in representing conditionally independent factors that generate stimuli and ignoring or abstracting away information related to conditionally dependent factors. The result of this representation formation strategy can, in the cases explored in this work, produce the commonly observed biases in human behavior.

The main limitation of the experiments in making more direct connections to participant representation formation is that they lacked data related to the eye gaze of participants, which could not be easily collected in an online platform like Mechanical Turk. Data from participant gaze location could answer the question of where the balance of task-relevance and reconstruction accuracy takes place, at the level of what information is chosen to attend to, or at the level of translating working memory to longer term internal representations that are useful for future recollection. This would be a relatively simple addition to this experiment paradigm and is an interesting area of future research.

Another important result is their relationship to the phenomenon of acquired equivalence, whereby dissimilar stimuli can be perceived as being more similar as a result of experience. While this phenomenon has generally been understood to be the result of differences in attention resulting from task relevance, the specific mechanisms of how representation formation can change as relevance is learned, and how this ultimately impacts representation formation, had not previously been well investigated.

The understanding provided by this model takes acquired equivalence to be an instance of a cognitive agent learning through experience that conditionally dependent factors w that are less relevant for memory formation $p(x'|z)$ but are relevant for performing an action $p(a|z)$, and thus must be represented in a disentangled manner in the latent generative factor z. The result is a representation that includes these previously dependent factors within the generative latent factors, which impacts the similarity of representations based on associations with similar experiences.

A key aspect of the proposed model and how it relates to biological perception is how it learns to retain information related to factors that are causally related to stimuli and to ignore or abstract away factors that are not. The participants in Experiment 1 were able to observe which factors were related to utility, which biased their change-detection accuracy. This was further explored in Experiment 2, where participants were required to learn through experience which factors were relevant, producing change-detection behavior that was initially not biased by stimuli utility, but that did eventually become biased as the factors relevant to the task were learned.

This understanding also provides insight into the source of differences in attentional allocation that can produce risk-sensitive behavior. In cases where areas of the visual scene that are more

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common are associated with lower utility outcomes, efficient encoding of that visual information allocates more resources onto faithfully reconstructing low utility information, which produces behavior that appears to be the result of more attention given to more common outcomes.

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(*Appendix follows*)

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Appendix

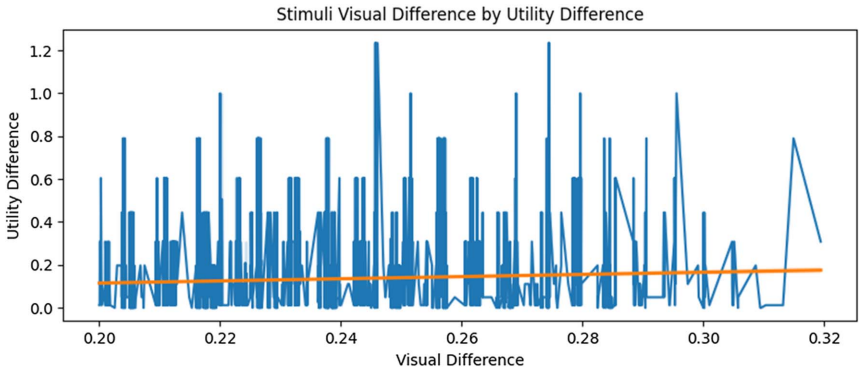
Stimuli Visual Difference Versus Utility Difference

Uβ-VAE Neural Network Parameters

The Uβ-VAEs neural network model used for experiments was pretrained on the full stimuli space with 1,000 epochs with a batch size of 100, and a mean utility for utility training. The encoder consisted of four convolutional layers, each with 32 channels, 4×4 kernels, and a stride of two. This was followed by two fully connected layers, each of 256 units. The utility output layers consisted of two fully connected layers each of 64 units. The image input size was (3, 64, 64). The latent distribution consisted of one fully connected layer of nine units parametrizing the mean and log standard deviation of 18 Gaussian random variables (or one unit in the first simulation example). The decoder architecture was simply the transpose of the encoder, but with the output parametrizing Bernoulli distributions over the pixels. Rectified Linear Unit activations were used throughout. An Adam optimizer was used for training with a learning rate of 5E-4.

Figure A1

Analysis of the Correlation of Visual Difference and Utility Difference of All Pairs of Possible Stimuli Used in Both Experiments



Note. This demonstrates that there is no clear correlation between visual differences and utility differences due to the large visual difference that occurs from rearranging the same set of marbles within a marble jar. See the online article for the color version of this figure.

Received June 27, 2023
Revision received May 22, 2024
Accepted June 9, 2024 ■

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